
Engineering Career Launch Program

An 8-week research & essay program for Theena

From dead-end jobs to a funded engineering path: eight weekly assignments that end with real applications submitted, a five-year plan, and an employer helping pay for the degree.

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How This Program Works

This guide is an 8-week program of research and essay assignments for an 18-year-old exploring mechanical or electrical engineering, designed to convert curiosity into a concrete plan: an entry-level job at an engineering firm, tuition reimbursement, and a degree path — instead of dead-end jobs.

The design behind the program

Each week produces a **real artifact she keeps** — a job list, a resume, a budget, an application actually submitted. The essays aren't busywork; each one forces her to process what she found and commit to a position. By week 8 she has:

1. A researched picture of what mechanical and electrical engineers (and technicians) actually do all day
2. A list of 15+ real, local, entry-level openings at engineering firms and manufacturers
3. A resume and cover letter targeted at those roles
4. A financial comparison: dead-end job vs. technician job + tuition reimbursement, in dollars
5. At least 2 informational interviews with people doing the work
6. At least 3 submitted applications
7. A written 5-year plan she authored herself

How to use this (a note for you, the parent)

- **One assignment per week.** Don't hand her this whole guide — it reads as a wall of homework instead of a project.
- **Let her present, not just submit.** The best version of this is a weekly 20-minute conversation where she walks you through what she found. The essay is her prep for that conversation.
- **Tie effort to real stakes, not just screen time.** The program works best when finishing an assignment unlocks something she wants, and when *she* starts seeing the payoff (Week 4's money math usually does this — the dollar gap between a dead-end job and a technician-plus-tuition path is genuinely startling).
- **Expect the goal to shift, and let it.** If her research convinces her that, say, an electrician apprenticeship or an engineering technology degree fits her better than a BS in engineering, that is the program *working* — she's making an informed decision instead of drifting. The win condition is "a researched plan she owns," not any one specific plan.

- **She should own the artifacts.** The job list, resume, and 5-year plan are hers. Your role by Week 8 is editor and cheerleader, not author.

Weekly overview

Week	Theme	Research deliverable	Essay
1	What do engineers actually do?	Comparison table: ME vs. EE, day-to-day reality	“Which of these lives do I actually want, and why?”
2	The hidden front door: technician & entry-level roles	List of 10 real job titles + what they pay locally	“The jobs nobody told me about”
3	Mapping the local market	15 real openings at real companies within commuting range	“Three companies I could see myself in”
4	The money: dead-end job vs. career path	Spreadsheet: 5-year earnings + tuition reimbursement math	“What my time is actually worth”
5	Education pathways	Comparison of community college, apprenticeship, and 4-year routes with real local costs	“My education plan and why I chose it”
6	Becoming a candidate	Resume + cover letter targeted at Week 3 openings	Cover letter (the essay IS the deliverable)
7	Talking to real people	2+ informational interviews completed, with notes	“What the people doing this job told me”
8	The plan	3+ applications submitted	“My five-year plan” (capstone)

Ground rules for every assignment

Written for Theena — one copy of these rules covers all eight weeks:

1. **Primary sources beat vibes.** Cite where every fact came from (job posting, BLS page, a person you talked to). “I read somewhere” doesn’t count.
2. **Real numbers, your area.** Salaries and tuition costs must be for your metro area, not national averages, wherever possible.
3. **Essays are 600-900 words** unless the assignment says otherwise. Long enough to think, short enough to finish.
4. **AI is a research assistant, not a ghostwriter.** Using AI to find sources or explain a concept is fine. The essay must be in your own words — the weekly conversation will make it obvious if it isn’t.

Part II — The Eight Weeks

Week 1 — What Do Engineers Actually Do All Day?

Goal: Replace the movie version of “engineer” with the real one, and figure out which flavor — mechanical or electrical — actually fits how you like to think and work.

Part 1: Research (aim for 3–4 hours across the week)

Build a comparison table for **mechanical engineering (ME)** and **electrical engineering (EE)** with these rows. Every cell needs a source.

Question	Mechanical	Electrical
What problems does this field solve? (3 concrete examples each)		
What does a typical Tuesday look like for a working engineer?		
What industries hire the most of them near me?		
Median pay in my metro area (entry level AND 10-year)		
What's the most math-heavy part of the job?		
What tools/software do they live in? (CAD? MATLAB? Oscilloscopes?)		
What do people in this field complain about?		

Where to look:

- U.S. Bureau of Labor Statistics *Occupational Outlook Handbook* (bls.gov/ooh) — search “mechanical engineers” and “electrical engineers.” Use the “Similar Occupations” links too.
- Reddit: r/MechanicalEngineering and r/ElectricalEngineering — search “day in the life” and “what I wish I knew.” Complaints are data.
- YouTube: search “day in the life mechanical engineer” (skip anyone selling a course).
- O*NET Online (onetonline.org) — the “Work Activities” section is the unvarnished truth.

Bonus research (optional, +respect): Find one engineer on LinkedIn or YouTube whose actual job sounds interesting to you. Write down their title, company, and what caught your attention. This becomes useful in Week 7.

Part 2: Essay (600–900 words)

Prompt: *“Which of these lives do I actually want, and why?”*

Requirements:

- Take a position. “Both seem cool” is not a position. It’s fine to be 60/40 — say so and say why.
- Use at least **4 specific facts** from your research table, cited.
- Include one honest paragraph about what worries you or seems hard about the field you’re leaning toward. (Pretending there are no downsides is how people pick wrong.)
- End with one question you can’t answer from the internet — something you’d have to ask a real engineer. Keep it; you’ll use it in Week 7.

Done when

- ☐ Comparison table complete, every cell sourced
- ☐ Essay written
- ☐ 20-minute walkthrough conversation done

Week 2 — The Hidden Front Door: Jobs Nobody Told You About

Goal: Discover the entry-level jobs that put you *inside* engineering companies without a degree — the roles that come with tuition reimbursement, mentors, and a ladder. These jobs mostly don't get advertised to 18-year-olds, which is exactly why most 18-year-olds end up in retail instead.

The big idea

There is a whole layer of jobs between “cashier” and “engineer”: **technicians, assemblers, drafters, test operators, quality inspectors, field service techs**. They pay more than retail, they teach you the actual work, the people around you are engineers, and many employers will pay for your degree while you work. Read Appendix A first — it's your starting vocabulary.

Part 1: Research

Build a list of **10 entry-level job titles** relevant to ME or EE. For each one:

1. **Job title** (as it actually appears on Indeed/LinkedIn — exact words matter for searching)
2. **What you'd do all day** (2–3 sentences, from real postings, not guesses)
3. **Typical pay in our area** (from actual local postings — record the range and the source posting)
4. **Requirements** (high school diploma? certificate? Which ones say “no experience necessary”?)
5. **Where it leads** (what does this job promote into?)

Where to look:

- Indeed / LinkedIn / ZipRecruiter — search the exact titles from Appendix A, filtered to your area
- Search tricks: "no experience" technician , "entry level" manufacturing , "will train"
- Local temp/staffing agencies that place in manufacturing (Aerotek, Adecco, Manpower) — their listings show what's really hiring

Rules:

- At least 5 of the 10 must be jobs you'd never heard of before this week.
- At least 3 must explicitly require no experience or say “will train.”

- Every pay number needs a link or screenshot of the posting it came from.

Part 2: Essay (600–900 words)

Prompt: *“The jobs nobody told me about.”*

Requirements:

- Pick the **3 titles from your list you’d most want**, ranked, and defend the ranking.
- Compare the best of them, honestly, to the jobs you’ve been taking: pay, what you learn, where each one is likely to put you in 3 years. Use real numbers.
- Address this question head-on: *“If these jobs are better, why do most people my age not get them?”* (Hint: it’s usually not ability.)

Done when

- ☐ 10-title list complete with sources
- ☐ Essay written
- ☐ Walkthrough conversation done

Week 3 — Mapping the Local Market

Goal: Turn last week’s job *titles* into a target list of real *openings* at real companies you could commute to. By the end of this week you’ll know your local engineering economy better than most adults in town.

Part 1: Research

A. Company census

Find **every company within your commuting range** that does mechanical or electrical engineering work: manufacturers, machine shops, electronics firms, utilities, HVAC/controls companies, defense contractors, medical device makers, automation/robotics integrators, engineering consultancies.

Target: **at least 12 companies**. For each:

- Name, location, rough size (LinkedIn shows employee count)
- What they actually make or do (one sentence)
- Whether their careers page mentions tuition reimbursement, apprenticeships, or “education assistance” (this matters for Week 4)
- Whether they mention **interns or co-op students** anywhere on the careers page — flag these companies; they matter later (see Appendix B)

Where to look:

- Google Maps: search “manufacturing,” “machine shop,” “electronics manufacturer,” “engineering” around your area and just scroll
- Your county or state’s Economic Development site — most publish a “major employers” list
- Local chamber of commerce member directory
- LinkedIn company search filtered by location + industry
- Drive past any industrial park you already know of and write down the names on the buildings — seriously, this works, and half those companies never post on job boards

B. Openings list

From those companies (plus Indeed/LinkedIn), collect **15 real, currently-open, entry-level positions** you could plausibly apply to. For each: title, company, pay (if listed), requirements, link, and application deadline if any.

Part 2: Essay (600–900 words)

Prompt: *“Three companies I could see myself in.”*

Requirements:

- Choose 3 companies from your census and go deep: what they make, who their customers are, what kinds of engineers they employ, what entry roles they have open or have posted recently.
- For each, answer: *what would I be able to learn here that I couldn't learn at a random job?*
- Rank them. Say what would make you pick #1 over #2.
- Note anything you found about tuition help at any of them, even a rumor to verify later.

Done when

- ☐ 12+ company census complete
- ☐ 15 real openings collected (links saved — postings expire, screenshot the good ones)
- ☐ Essay written
- ☐ Walkthrough conversation done

Keep the openings list safe. Weeks 6 and 8 are built directly on it.

Week 4 — The Money: What Your Time Is Actually Worth

Goal: Do the math, in real dollars, on two versions of the next five years: (A) drifting between dead-end jobs, and (B) an entry-level technician job at an engineering firm with tuition reimbursement. This is the week the whole plan usually stops being your parent's idea and starts being yours — or doesn't. Either way, you'll know.

Part 1: Research + spreadsheet

Build a spreadsheet (Google Sheets is fine) comparing two 5-year paths:

Path A: The default

- Use the actual hourly wage of the kind of jobs you've been taking
- Realistic raises for that kind of work (look up what a shift lead makes — that's usually the ceiling)
- Where does Path A leave you at 23? (wage, skills, degree progress: none)

Path B: Technician + tuition reimbursement

- Starting wage: use a real number from your Week 3 openings list
- Typical technician raises (postings for "Technician II" / "Senior Technician" in your area tell you this)
- Tuition reimbursement: research what employers actually offer. Read Appendix C, then **verify current numbers** for at least 3 employers from your Week 3 census — programs change, so the number you use must come from the company's current careers page or a current posting
- Cost of your local community college per year + your state university's engineering program per year (in-state, from their actual tuition pages)
- How much of that does reimbursement cover? What's left over, and how would you cover the gap?

The comparison

- Total 5-year earnings, Path A vs Path B
- Degree progress at 23 on each path
- Estimated debt at 23 on each path
- What each path's year 6 looks like (this is where the lines really split — an engineering graduate's starting salary vs. a shift lead's wage)

Sources: real postings, real tuition pages, BLS for wage trajectories. Every number in the sheet gets a source note.

Part 2: Essay (600–900 words)

Prompt: *“What my time is actually worth.”*

Requirements:

- Present your findings like you’re the analyst and the reader is the client. Lead with the bottom line: what is the total 5-year gap in dollars?
- Steelman Path A: what are the *real* reasons people stay on it? (Flexibility? Low stress? Familiarity? Immediate cash?) Take those reasons seriously, then respond to them.
- Be honest about Path B’s costs: working while studying is hard. What would it actually demand from you, and do you believe you’d pay it?
- End with a verdict. Yours, not the spreadsheet’s.

Done when

- ☐ Spreadsheet complete, every number sourced
- ☐ Tuition benefits verified from at least 3 current employer sources
- ☐ Essay written
- ☐ Walkthrough conversation done

Week 5 — Education Pathways: Picking Your Route

Goal: Choose, with evidence, how you’d actually get the education — because “go to school for engineering” hides at least four very different routes with very different costs, timelines, and lifestyles. Read Appendix D first for the map; this week you fill in the local details.

Part 1: Research

Investigate all four routes **as they actually exist where you live:**

Route 1: Community college → transfer to a 4-year engineering program

- Which community college(s) near you have an engineering or pre-engineering track?
- Do they have a **transfer agreement (articulation agreement)** with your state university’s engineering school? Find the actual agreement page.
- Cost per year, class schedule options (evening? online? — matters if you’re working)

Route 2: Engineering Technology degree (AS or BS in MET/EET)

- What’s the difference between *engineering* and *engineering technology*? (Look up both; ask r/AskEngineers)
- Who near you offers it, what does it cost, and what jobs does it lead to?

Route 3: Apprenticeship

- For EE-leaning: IBEW electrician apprenticeships (paid from day one, “earn while you learn”). Find your local IBEW chapter and their application requirements/timeline.
- For ME-leaning: machinist/toolmaker/industrial maintenance apprenticeships — check apprenticeship.gov for registered programs near you.
- What do apprentices earn in year 1 and at journeyman level in your area?

Route 4: Straight to a 4-year university

- In-state cost, admission requirements, and what the first two years actually look like (hint: the same calculus/physics you’d take at community college for a third of the price)

For each route, record: total cost, years to a credential, whether you can work while doing it, whether ABET accreditation applies (see Appendix D for why you should care), and what door it opens at the end.

For every college on your list, also ask (read Appendix B first — enrollment is the key that unlocks internships and co-ops):

- Do you have a **co-op program**? Which local employers take your students?
- Do students get **Handshake** access, and does the career center place pre-engineering/technology students?
- Which employers sponsor or hire your engineering-track students? (The college knows exactly who.)

Part 2: Essay (700–1000 words)

Prompt: *“My education plan and why I chose it.”*

Requirements:

- Commit to a route (combinations count — e.g., “technician job + community college nights + transfer” is a route).
- Show your work: why this one over the other three? Use your cost and timeline numbers.
- Name the **first three concrete steps** and put a date on each (e.g., “meet with the community college advisor before ,” ***“IBEW application window opens”***).
- Name the biggest risk to your plan and what you’d do if it happens.

Done when

- ☐ All four routes researched with local specifics
- ☐ Essay written with dated next steps
- ☐ Walkthrough conversation done

Week 6 — Becoming a Candidate

Goal: Build the resume and cover letter that get you one of the Week 3 jobs. This week the essay *is* the deliverable — a cover letter is just a persuasive essay with a job on the line.

Part 1: The resume

Build a one-page resume aimed at the technician/entry-level roles on your Week 3 list.

The reframe that makes this work: you are not “an 18-year-old with random jobs.” Those dead-end jobs are raw material. Mine them:

- Showed up on time for X months → *reliability* (genuinely rare; hiring managers care)
- Handled a register / inventory / closing → *accountable for money and process*
- Dealt with angry customers → *stays calm under pressure, communicates clearly*
- Trained a new hire → *leadership, even a little*
- Any fixing, building, assembling, crafting, car work, 3D printing, soldering, gaming-PC building → *this is mechanical/electrical aptitude and it goes ON the resume*

Requirements:

- One page. Clean layout, no photo, no graphics, PDF.
- A 2–3 line summary at the top that says what you’re aiming at: e.g., “*Reliable, mechanically-inclined worker seeking an entry-level technician role while pursuing an engineering degree.*” (Employers with tuition programs *like* seeing the degree goal — you’re exactly who those programs are for.)
- Tailor the skills section to words that actually appear in your Week 3 postings (this is how resume screening works — mirror their vocabulary honestly).
- Get one adult who has hired people to tear it apart, then revise.

Part 2: Cover letter (this week’s essay, 300–400 words)

Pick the **#1 job** on your Week 3 list and write a real cover letter for it.

Structure that works: 1. **Why this company** — one specific thing about what they make/do that genuinely interests you (Week 3 research pays off here; almost no applicant does this) 2. **Why you** — two concrete proof points from your experience (see reframes above) 3. **Where you’re going** — your degree plan, and why that makes you a better hire, not a flight risk: you’re building a career in their industry, and if they have tuition assistance you intend to earn it 4. **Ask for the interview.** Plainly.

Then write a second version for your #2 job. You'll feel the difference between tailored and generic — that difference is the whole game.

Done when

- ☐ One-page resume done, reviewed by someone who has hired people, revised
- ☐ Two tailored cover letters written
- ☐ Walkthrough conversation done — read the cover letters out loud; if a sentence sounds like nobody would ever say it, rewrite it

Week 7 — Talking to Real People

Goal: Have real conversations with at least **two people** who do the work — an engineer, a technician, an apprentice, a community college engineering instructor, anyone actually on the path. This is the single highest-value week and the one that most requires leaving your comfort zone. That's not a side effect; it's the point. Interviews, first days, and asking for raises all live outside the comfort zone too, and this is the practice version where nothing bad can happen.

Part 1: Get the conversations

Finding people (in rough order of ease): 1. **Family network first.** Parents' friends, friends' parents, neighbors, anyone at church/gym/etc. who works in manufacturing, utilities, construction trades, or engineering. One ask — “do you know anyone who works as an engineer or technician I could talk to for 20 minutes?” — usually produces someone. 2. **Community college.** Email an engineering/engineering-tech instructor: you're deciding between programs and would love 15 minutes. Instructors answer these emails; recruiting interested students is their job. 3. **LinkedIn.** Find technicians/engineers at your Week 3 companies. Short message: who you are, one specific thing about their company from your research, and a request for 15-20 minutes of advice. Send 8-10 to get 2 yeses — low reply rates are normal and not about you. 4. **Walk in.** Small machine shops and repair shops are often genuinely happy to give a curious 18-year-old ten minutes and a tour. Go at a non-busy hour, be brief, be polite.

The magic words: people flinch at “can I interview you” but say yes to *“I'm trying to figure out if this is the right path for me — could I ask you a few questions about how you got started?”* You are asking for advice, not a job. People love giving advice.

Prepare 6-8 questions. Good ones:

- How did you get your start? What would you do differently at 18?
- What does a normal day actually look like?
- What separates the new hires who move up from the ones who don't?
- Does your company help pay for school? How does that work in practice — do people really use it?
- What should I learn *right now* to be useful on day one?
- Does your company take **interns or co-op students**? Who runs that program?
- Your held-over question from Week 1 — the one the internet couldn't answer
- End with: *“Is there anyone else you think I should talk to?”* (This is how 2 interviews become 4.)

Afterward: send a thank-you text/email within 24 hours. Every time. Write notes immediately — you'll forget the good stuff within a day.

Part 2: Essay (600–900 words)

Prompt: *“What the people doing this job told me.”*

Requirements:

- What did each person say that you couldn't have gotten from the internet?
- What surprised you? What contradicted your Weeks 1–5 research? (When a person and a website disagree, that's the most interesting data you have — dig into why.)
- Did anything change your plan? Even “no, and here's why I'm more sure now” counts — but defend it.
- Be honest about the hardest part: reaching out. What did the discomfort feel like before vs. after? Was the anticipation worse than the conversation? (It almost always is. Notice that — it's the most transferable finding of the whole program.)

Done when

- ☐ 2+ real conversations completed, notes written up
- ☐ Thank-you messages sent
- ☐ Essay written
- ☐ Walkthrough conversation done

Week 8 — The Plan (Capstone)

Goal: Ship it. Applications actually submitted, and a five-year plan in writing that is *yours* — built from seven weeks of your own research, your own math, and real conversations. Nobody can assign you a future; this week you write your own.

Part 1: Apply

Submit **at least 3 real applications** from your Week 3 list (refresh it first — postings from five weeks ago may be gone, and new ones will be up).

For each application:

- Tailor the resume’s skill words and the cover letter to the posting (Week 6 taught you how)
- Record it: company, title, date applied, contact if any, follow-up date
- **Follow up after ~1 week** if you hear nothing: short, polite email or call restating interest. Most applicants never do this; it visibly separates you.

Also do at least one of:

- Ask one Week 7 contact if they’d refer you or pass your resume along (referrals beat cold applications by a mile — this is why Week 7 came first)
- Register with one staffing agency that places in local manufacturing
- Book an actual appointment with a community college advisor or an apprenticeship program

Part 2: Capstone essay (1000–1500 words)

Prompt: *“My five-year plan.”*

This is the document you’ll reread at 20 and 23. Requirements:

1. **Where I’m going:** the role/credential you’re aiming at by ~23, and why — grounded in what you learned, citing your own earlier work (“my Week 4 spreadsheet showed...”, “the technician I interviewed said...”)
2. **The route:** job + education path, with the money math (what you’ll earn, what school costs, what reimbursement covers, what you’ll cover)
3. **The next 90 days:** specific and dated. Applications, follow-ups, advisor meetings, application windows, deadlines. Include your **first internship or co-op window:** the first term you’ll be eligible (enrolled), and the date those applications open — big-company summer internships open the previous August–October (see Appendix B).

4. **Milestones:** what should be true at 6 months, 1 year, 2 years, 5 years — concrete enough that future-you can check them (e.g., “employed as a technician within 4 months,” “12 college credits by next summer”)
5. **Failure modes and answers:** the three most likely ways this plan goes sideways (no offers? class + work overload? interest fades?) and what you’ll actually do in each case
6. **What I’d tell myself from eight weeks ago.** One honest paragraph.

Part 3: Present it

Present the plan out loud — 15 minutes, like a real review. You’re not asking for permission; you’re briefing your stakeholders. Take questions.

Done when

- ☐ 3+ applications submitted and logged, follow-up dates set
- ☐ One referral/agency/advisor action taken
- ☐ Capstone essay written
- ☐ Plan presented

After Week 8

The program ends; the plan doesn’t. The tracker’s application log keeps going until there’s an offer. And the plan should get a 90-day review on the calendar — same format, 15 minutes: what happened, what’s next.

Appendix A — Entry-Level Jobs That Lead Into Engineering

The job titles below are the “hidden front door”: roles at engineering firms, manufacturers, and utilities that hire people with a high school diploma, teach real skills, pay better than retail, and often come with tuition assistance. Use these **exact phrases** when searching Indeed/LinkedIn — job boards match words.

Mechanical-leaning

Title (search these words)	What it is	Typical entry requirement
Manufacturing / Production Technician	Run and monitor production equipment, basic troubleshooting	HS diploma, will train
Mechanical Assembler / Electro-Mechanical Assembler	Build products from drawings — reading blueprints is the skill	HS diploma, will train
CNC Operator (→ CNC Machinist)	Load/run computer-controlled machines; machinist is the skilled version	HS diploma; operator roles often train
Quality Inspector / QC Technician	Measure parts with calipers/micrometers/CMM, check against specs	HS diploma, detail orientation
Maintenance Technician (Industrial)	Keep plant machinery running — mechanical + electrical mix, very promotable	HS diploma or short certificate
Test Technician	Run test procedures on products, log results, flag failures	HS diploma, will train
HVAC Installer / Technician Helper	Heating/cooling systems — thermodynamics with your hands	HS diploma; helper roles train
Drafter / CAD Technician (junior)	Turn engineers' sketches into CAD drawings	Certificate or a semester of CAD — a community college CAD class makes this reachable fast

Electrical-leaning

Title (search these words)	What it is	Typical entry requirement
Electronics Technician	Assemble, test, repair circuit boards and devices; soldering	HS diploma or certificate
Electrical Assembler / Wire Harness Assembler	Build wiring assemblies from diagrams	HS diploma, will train
Test Technician (Electrical)	Bench-test electronics with multimeters, oscilloscopes	HS diploma, will train
Field Service Technician (junior)	Install/repair equipment at customer sites	HS diploma + driver's license
Electrician Apprentice / Helper	The classic earn-while-you-learn path; IBEW apprenticeships are paid from day one	HS diploma + algebra; union programs have an application process
Panel Builder / Control Panel Assembler	Wire industrial control cabinets from schematics	HS diploma, will train
Utility Worker / Meter Technician	Entry roles at power/water utilities — utilities have strong benefits and tuition programs	HS diploma
Instrumentation Technician Helper	Sensors and control systems in plants	HS diploma or certificate

Either / both

- **Engineering Aide / Engineering Technician (junior)** — direct support to engineers: measurements, documentation, prototypes. The single most on-ramp title there is; less common but always worth searching.
- **Lab Technician (materials / calibration)** — testing labs hire entry-level and teach precision.
- **Warehouse → Maintenance pipeline** — big distribution centers (Amazon, UPS, big-box logistics) promote reliable people into maintenance tech roles *and* have well-known tuition programs. A warehouse job **at a company with a real education benefit and a maintenance-tech ladder** is categorically different from a dead-end warehouse job — the difference is whether you're using the ladder.

What makes any of these NOT a dead end

A job is a dead end or a launchpad depending on four things. Check every posting against this list:

1. **Ladder:** is there a visible Tech I → Tech II → Senior/Lead progression?
2. **Skills:** will you leave knowing things the next employer pays for (blueprint reading, soldering, CAD, PLCs, calibration)?
3. **Education benefit:** tuition reimbursement/assistance, and do people actually use it?
4. **Proximity to engineers:** will you work near the people whose job you want? (You can't get pulled up a ladder you're not standing next to.)

Three or four checks = launchpad. Zero or one = it's retail with extra steps.

Appendix B — Internships, Co-ops, and How to Actually Get One

Internships and co-ops are the highest-octane version of this whole plan: paid, credentialed, engineering-adjacent work that converts directly into full-time offers. This guide covers what they are, the one gate that controls access to most of them, and the paths that are open **right now** to a high school graduate.

The vocabulary

Term	What it means	Typical length	Paid?
Internship	A temporary position, usually summer, doing junior-level real work	10–12 weeks	Engineering ones almost always pay (often \$18–30+/hr)
Co-op	A formal partnership between a college and employers: you alternate full-time school semesters and full-time paid work semesters, or work part-time year-round	3–12 months per rotation, often multiple rotations	Yes, and often better than internships
Pre-apprenticeship	A short program (weeks to a few months) that preps you for and feeds into a registered apprenticeship	4–16 weeks	Sometimes paid, usually free to attend
Youth/registered apprenticeship	Paid job + structured training from day one (see Appendix D)	1–5 years	Yes, from day one

The gate nobody tells you about

Most engineering internships and co-ops require you to be a currently enrolled student. Not a graduate, not “planning to enroll” — enrolled. It’s how companies filter, it’s how their intern programs are legally structured, and it’s why a motivated 18-year-old applying to “Engineering Intern — Summer 2027” postings with no enrollment gets auto-rejected while a mediocre sophomore gets an interview.

The move this implies: enrolling at community college — even one or two evening classes while working — flips the switch. The day you have a student ID and an expected graduation date:

- You can put “Pre-engineering student, [College], expected [year]” on your resume

- You qualify for **Handshake** (the college-recruiting platform nearly every school uses — ask the college’s career center for access; local employers post there specifically to find students like you, with far less competition than Indeed)
- The college’s **career center and co-op office** start working for you for free — ask directly: “which local companies take co-op students from here?” They keep a list.
- Engineering internship postings stop auto-rejecting you

This is worth saying plainly: **one \$200 evening class can unlock a \$25/hr summer internship.** Enrollment is not just degree progress — it’s a key.

What’s open BEFORE enrollment (right now, with a HS diploma)

1. **Registered apprenticeships and pre-apprenticeships** — no enrollment needed; apprenticeship.gov by ZIP, IBEW locals, and community colleges (ask who sponsors “mechatronics” or “advanced manufacturing” students).
2. **Manufacturing summer/seasonal hires at local firms** — small manufacturers staff up seasonally; a summer as a production helper at an engineering firm IS an internship in everything but the title. Week 3’s company census is the target list.
3. **The self-made internship** (see the pitch below) — small companies don’t have intern *programs*, which means they also don’t have intern *requirements*. The owner can just say yes.
4. **State workforce programs** — many states run paid youth/young-adult work programs (search “[your state] youth apprenticeship” and “[your state] workforce development young adult”); your local American Job Center (careeronestop.org finds it) can enroll you in some of these for free.
5. **A regular technician job** — never forget that the Week 2/3 technician path *is* the working person’s internship: paid, full-time, and it doesn’t end in August.

The self-made internship pitch

Small machine shops, controls integrators, and repair outfits (5–50 people) rarely post internships but often say yes to this, in person or by short email, addressed to the owner or shop manager:

“I’m starting an engineering degree at [college] in [month]. Before that, I want real shop experience more than I want a mall job. Would you take me on for the summer — sweeping, deburring, organizing stock, whatever’s useful — so I can learn how a real shop runs? I’m reliable and I’ll show up every day.”

Why it works: it asks for the bottom rung explicitly, signals reliability (their actual biggest hiring problem), and has an end date so it’s low-risk for them. Ten asks will get a yes. That

yes goes on the resume as *“Shop Assistant, [Company] — supported machinists in a production environment.”*

Once enrolled: playing the internship game properly

- **Timeline is everything.** Summer engineering internships at big companies open applications the **August-October before** and are largely filled by February. Mark the calendar; “I’ll apply in spring” means “next year.”
- **Where:** Handshake first (least competition, local employers), then company careers pages directly (defense contractors, utilities, and manufacturers from the Week 3 census all run intern programs), then Indeed/LinkedIn with “engineering intern” + your area.
- **Community college students CAN get these.** Big employers (especially defense and semiconductors) explicitly recruit community college and engineering-tech students for technician internships. The GPA + a technician work history from this program beats a university sophomore’s empty resume more often than you’d think.
- **Co-op vs. internship when transferring:** when Week 5’s college research happens, ask each program: “do you have a co-op program, and which employers take your students?” A school with a real co-op pipeline is worth ranking higher — co-op students routinely graduate with 12+ months of paid engineering experience and a standing job offer.
- **The referral shortcut still rules.** A Week 7 contact saying “we take interns, send me your resume” skips the entire pile. Every informational interview should end with: *“Does your company take interns or co-op students? Who runs that?”*

How this fits the 8-week program

- **Week 3:** when building the openings list, also flag any company that mentions interns, co-ops, or apprentices anywhere on its careers page.
- **Week 5:** ask every college about co-op programs and Handshake access; ask the community college which employers sponsor students.
- **Week 7:** add the intern question to every interview (wording above).
- **Week 8:** the five-year plan should name the first internship/co-op window you’ll be eligible for and the date applications open for it.

Appendix C — Employers Known for Paying for School

How to use this list: it's a starting map, not gospel. Programs change yearly — before using any number in your Week 4 spreadsheet, **verify it on the company's current careers/benefits page or in a current job posting**. Also: local mid-size manufacturers you found in Week 3 often have tuition benefits that never make lists like this — the only way to find out is to check their careers page or ask in an interview (“does the company offer education assistance?” is a great question that also signals ambition).

Useful background

- The IRS lets employers provide up to **\$5,250/year in tax-free tuition assistance** (IRC Section 127). That's why so many programs land at exactly that number.
- “Tuition reimbursement” (you pay, they pay you back per grade/semester) is different from “tuition paid up front” — up-front programs are much easier when you don't have savings. Note which kind each employer offers.
- Many programs restrict field of study to something job-related — engineering almost always qualifies at these employers, but confirm.
- Ask about **strings**: minimum tenure before eligibility (often 90 days–1 year), GPA requirements, and payback clauses if you quit soon after.

National employers with well-known education programs (verify current details)

Directly in the engineering/industrial world

- **Defense & aerospace:** Boeing, Lockheed Martin, Raytheon (RTX), Northrop Grumman, General Dynamics, BAE Systems — historically strong tuition programs and huge technician workforces. They hire entry-level assemblers and technicians.
- **Industrial/manufacturing:** GE (Aerospace/Vernova), Siemens, Honeywell, Caterpillar, John Deere, 3M, Emerson, Eaton, Schneider Electric.
- **Semiconductors/electronics:** Intel, Micron, Texas Instruments, GlobalFoundries — fabs hire technicians with HS diplomas and are famous for funding associate-degree pipelines with local colleges.
- **Utilities:** almost every regional electric/gas utility has education assistance and technician apprenticeships. Look up *your* utility's careers page.
- **Automotive:** Toyota, Honda, and other manufacturers with US plants; also dealership networks' technician programs.

Not engineering jobs, but famous education benefits (bridge options)

These can fund school while you apply toward technical roles, and some have internal ladders into maintenance/technical work:

- **Amazon (Career Choice)** — funds degrees/certificates including at community colleges; has a maintenance-tech ladder internally
- **UPS (Earn & Learn)** — part-time package handlers get tuition assistance; part-time-friendly for full-time students
- **Home Depot / Lowe's** — tuition programs; store roles also build real product knowledge in electrical/building trades
- **Starbucks, Chipotle, Target, Walmart** — well-known college programs (often through specific partner schools — check whether *your* school qualifies)

A bridge job at one of these is Plan B, not Plan A: Plan A is a technician job that teaches the work *and* helps with school. But a bridge job with a real education benefit still beats a dead-end job with none.

Apprenticeships: the other way employers pay for education

- **IBEW/NECA electrician apprenticeships** — you're paid to work while the program covers your classroom training; journeyman electricians earn a genuinely good living, and the credential is portable everywhere. Find your local at ibew.org.
- **Registered apprenticeships** in machining, industrial maintenance, mechatronics: search apprenticeship.gov by ZIP code.
- Some manufacturers run their own apprentice programs with local community colleges (often called “mechatronics” or “advanced manufacturing” partnerships — ask the community college which local employers sponsor students; the college knows exactly who pays for what).

Verification checklist (per employer, for Week 4)

- ☐ Current benefit found on the company's own site or a ≤ 6 -month-old posting
- ☐ Amount per year, and reimbursement vs. paid-up-front
- ☐ Eligibility wait, hours threshold (full-time only? 20+ hrs?)
- ☐ Field-of-study restrictions
- ☐ Payback clause if you leave
- ☐ Bonus: an employee's firsthand account (Reddit, or your Week 7 interviews) that people actually use it

Appendix D — Education Pathways Into Engineering

The map behind Week 5. Four routes, honestly compared.

Route 1: Community college → transfer (the working person's classic)

Two years of the same calculus, physics, chemistry, and intro engineering at a community college — typically at a third (or less) of university cost — then transfer into a university engineering program for the final two years.

- **Why it fits this plan:** evening/online sections exist, so it stacks with a full-time technician job; tuition reimbursement often covers most or all of community college cost.
- **The one thing you must verify:** a formal **articulation/transfer agreement** between the community college and the target university's *engineering school* (not just the university generally). Engineering programs are picky about which math/physics/intro courses transfer. Meet an advisor at BOTH schools and get the course plan in writing.
- **Watch out for:** taking classes that don't transfer (the advisor meeting prevents this), and the transfer GPA bar for engineering admission.

Route 2: Engineering Technology (MET/EET)

A different degree: **engineering technology** (2-year AS or 4-year BS) is more applied and hands-on, less theoretical math. Graduates become technologists, senior technicians, manufacturing/test/field engineers.

- **Honest trade-off:** a BS in Engineering Technology is *not* interchangeable with a BS in Engineering. Some employers treat them the same for many roles; some don't. Licensure (PE) with an ET degree is possible in some states, harder in others. If the dream is "design engineer at an aerospace firm," lean Route 1/4. If it's "well-paid hands-on technical career, working sooner," ET is excellent and *underrated*.
- **Fits working students well** — programs are often at community colleges and regional universities with evening options, and coursework connects directly to a technician day job.

Route 3: Apprenticeship (earn from day one)

Paid work + paid classroom training → journeyman credential in ~4-5 years, no debt, solid income the whole way. Electrician (IBEW), machinist, industrial maintenance, mechatronics.

- **Honest trade-off:** it's a trades credential, not an engineering degree — a different (excellent) career, not a slower route to the same one. Plenty of people later stack a degree on top (some locals/employers help pay), and journeyman experience makes a later EE/ME student dramatically better at the job.
- **Reality check:** good apprenticeships are competitive with application windows and aptitude tests (algebra matters). Missing a window can mean waiting a year — find the dates *early*.

Route 4: Straight to a 4-year university

The default path. Four (realistically often five) years, highest sticker price, full college experience, engineering internships between years.

- **When it wins:** strong financial aid, scholarships, or family funding; or admission to a program whose brand/co-op pipeline genuinely changes outcomes.
- **When it loses:** paying university prices for the freshman/sophomore courses community college sells for a third the price, funded by loans instead of an employer.

ABET, in one paragraph

ABET accreditation is the quality stamp for engineering and engineering technology programs. It matters because: (1) the FE/PE licensure track expects an ABET-accredited degree, (2) many employers (especially defense) require it, and (3) transfer agreements are built around it. Before committing money to ANY program, search it on abet.org. Community colleges themselves usually aren't ABET-accredited (that's normal — the *destination* program is the one that must be), but pre-engineering courses must be transferable to an ABET program.

The combination most likely to fit this plan

Technician job (full-time) + community college (evenings) + employer tuition money → transfer or ET completion. Slower per-year than full-time school; graduates at ~23-24 with experience, income history, industry contacts, and little or no debt — usually *ahead* of the classmate who went straight through, in everything except graduation date. But Week 5's job is to check whether that's true *for you, where you live* — not to take this file's word for it.

Appendix E — Essay Rubric

Score each essay 1–4 on five dimensions (max 20). Share the rubric *before* she writes — a rubric is a target, not a gotcha. **17+ = excellent. 13–16 = solid. Below 13 = one focused revision pass on the weakest dimension**, not a redo of the whole essay.

The five dimensions

1. Evidence (did she actually do the research?)

- **4** — Specific facts, local numbers, named sources woven through; you could check her claims
- **3** — Real research present, but some claims float without sources
- **2** — Mostly general statements that could have been written without doing the research
- **1** — Vibes

2. Position (did she commit to a point of view?)

- **4** — Clear stance, defended; addresses the strongest counterargument honestly
- **3** — Takes a stance but dodges the hard opposing points
- **2** — Fence-sitting; “both have pros and cons” with no verdict
- **1** — No discernible position

3. Honesty (does it engage with the hard parts?)

- **4** — Names real doubts, costs, and risks, and reasons through them; sounds like a person deciding, not a brochure
- **3** — Acknowledges downsides but hand-waves them
- **2** — Tells the reader (you) what they want to hear
- **1** — Pure performance

4. Her own thinking (is anything here *hers*?)

- **4** — Draws her own connections and conclusions; at least one idea that wasn’t in any source
- **3** — Good synthesis of sources but nothing beyond them
- **2** — Assembled summary of what she read
- **1** — Reads like AI or a template filled in

5. Craft (is it readable and finished?)

- **4** — Clear structure, clean sentences, within word count, proofread
- **3** — Readable with rough patches
- **2** — Disorganized or clearly unproofread
- **1** — Fragmentary

Notes for the grader (you)

- **Grade the thinking, not the conclusion.** If she argues well for a path you disagree with, that's a 4 on Position — the program's goal is that she can research and decide, not that she reaches a pre-approved answer. Argue back in the conversation, as an equal, if you disagree.
- **The conversation is half the assessment.** Someone who did the work can go off-script, answer "what surprised you?", and defend choices. Someone who didn't, can't. If the essay is thin but the conversation is rich, the learning happened — say so, and coach the writing separately.
- **Praise Honesty (dimension 3) loudest.** It's the dimension teenagers sandbag first when writing for a parent, and the whole program depends on her telling you what she really thinks.
- **Feedback ratio:** name the two strongest things before the one thing to fix. You want more essays, which means writing them has to feel like winning.

Part IV — Keeping Score

Program Tracker

Weekly progress

Week	Assignment	Research done	Essay done	Conversation done	Rubric score	Notes
1	What engineers actually do	☐	☐	☐		
2	The hidden front door	☐	☐	☐		
3	Mapping the local market	☐	☐	☐		
4	The money	☐	☐	☐		
5	Education pathways	☐	☐	☐		
6	Becoming a candidate	☐	☐	☐		
7	Talking to real people	☐	☐	☐		
8	The plan (capstone)	☐	☐	☐		

Artifacts produced (the stuff that outlives the program)

- ☐ ME vs. EE comparison table (Week 1)
- ☐ 10 entry-level job titles with local pay (Week 2)
- ☐ Company census: 12+ local engineering employers (Week 3)
- ☐ 15 real openings list (Week 3)
- ☐ 5-year money spreadsheet (Week 4)
- ☐ Education route comparison with local costs (Week 5)
- ☐ One-page resume, reviewed and revised (Week 6)
- ☐ 2 tailored cover letters (Week 6)
- ☐ Interview notes from 2+ real conversations (Week 7)
- ☐ Five-year plan, presented (Week 8)

Application log (keeps going after Week 8)

Date applied	Company	Title	How (site/referral/agency)	Follow-up date	Status

Contact log (Week 7 and beyond)

Name	Role / company	How we met	Last contact	Owes / owed

90-day review (after Week 8)

Date scheduled: _____

- What's the job status?
- What's the school status (advisor met? enrolled? application windows)?
- What in the five-year plan changed, and why?
- Next 90 days' three concrete steps: